Fiscal Year:	FY 2016	**Task Last Updated:**	FY 03/24/2016
PI Name:	Yodh, Arjun Ph.D.		
Project Title:	Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	06/10/2013	**End Date:**	06/09/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AL27G		
Performance Goal No.:			
Performance Goal Text:			
	This proposal builds on and continues the research described in Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2) (Ed. note: see "Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2)"; grant NNX08AO80G in Task Book). The roots of the work derive from an experimental program that was peer-reviewed and accepted for flight on the International Space Station (ISS); the instrumentation at the heart of the original flight proposal for ISS was part of the so-called Light Microscopy Module (LMM). More recently, a Science Concept Review (SCR) for Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2) was held in June 2011, and the proposed research was strongly endorsed by the External Review Panel in their report on June 14 2011; thus, the science goals and experimental plans for the present proposal, Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3), have already been critically reviewed. The current research will be integrated into the hardware developed for the space-flights that are named by variants of the term “ACE” (e.g., ACE-H1). In addition, related complex fluid measurements have been carried out in the Binary Colloid Alloy Test (BCAT) BCAT-series flight experiments. Since		

Task Description:	the original proposal Low Volume Fraction Entropically Driven Colloidal Assembly was submitted and approved, the NASA budget has been in flux and soft matter science has evolved too. In response to these developments, the science aims have evolved to take advantage of important recent breakthroughs in the field. Herein we describe these complex fluid experiments. Note also, whenever possible, we will be joining forces with our colleagues from Europe to perform collaborative experiments within the European Space Agency (ESA) and NASA facilities. The overall rationale for these investigations is that the elimination of particle sedimentation effects in microgravity creates a purely “thermodynamic” environment for colloidal suspensions wherein particle size, shape, and volume fraction, and interparticle interactions are the dominant determinants of the assembled structures. This feature enables us to make new materials that do not form on Earth and to elucidate the effects of gravity in soft matter science that are masked in ground-based experiments. The Phase 3 version of Low Volume Fraction Entropically Driven Colloidal Assembly explores a range of experimental complex fluids with themes related to and evolved from previous proposal aims. The original studies were centered around entropically driven low volume-fraction self-assembling complex fluids, suspensions composed of an ensemble of large particles, and additional smaller constituents (either a small colloidal particle, a polymer, or a micelle). The present studies retain many of the original themes but also have new focus that reflects the evolution of the complex fluid field, as well as new developments in our laboratory. Broadly, the proposed research will elucidate a wide-range of phenomena in condensed matter physics that has implications for phase-diagrams of entropically driven colloidal media, crystallization, and melting mechanisms in three-dimensional (3D) bulk media and in finite volumes, jamming, and glass formation in 3D, and, opportunistically, crystallization and melting mechanisms and aging of colloids/soft-matter based on more anisotropic particles and filaments on a variety of length scales. International collaborators are Prof. Luca Cipelletti, Université Montpellier II, Montpellier, France; Prof. Roberto Piazza, Department of Nuclear Engineering, Politecnico di Milano, Milan, Italy; Prof. Willem Kegel, Van 't Hoff Laboratory for Physical and Colloid Chemistry, Debye Research Institute, Utrecht University, Utrecht, The Netherlands.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research can provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.
Task Progress:	The abstracts that follow summarize the publications from January, 2015 through April, 2016 that are the result of support from the NASA grant and acknowledge award number NNX13AL27G. Corresponding citations are found in the Task Book Bibliography section below: An experimental and theoretical study of lyotropic chromonic liquid crystals (LCLCs) confined in cylinders with degenerate planar boundary conditions elucidates LCLC director configurations. When the Frank saddle-splay modulus is more than twice the twist modulus, the ground state adopts an inhomogeneous escaped-twisted configuration. Analysis of the configuration yields a large saddle-splay modulus, which violates Ericksen inequalities but not thermodynamic stability. Lastly, we observe point defects between opposite-handed domains, and we explain a preference for point defects over domain walls. This study introduces liquid crystal (LC) Janus droplets. We describe a process for the preparation of these droplets, which consist of nematic LC and polymer compartments. The process employs solvent-induced phase separation in emulsion droplets generated by microfluidics. The droplet morphology was systematically investigated and demonstrated to be sensitive to the surfactant concentration in the background phase, the compartment volume ratio, and the possible coalescence of multiple Janus droplets. Interestingly, the combination of a polymer and an anisotropic LC introduces new functionalities into Janus droplets, and these properties lead to unusual dynamical behaviors. The different densities and solubilities of the two compartments produce gravity-induced alignment, tumbling, and directional self-propelled motion of Janus droplets. LC Janus droplets with remarkable optical properties and dynamical behaviors thus offer new avenues for applications of Janus colloids and active soft matter. A simple method to prepare temperature-sensitive films composed of micrometer-sized colloidal hydrogel particles using evaporating drops of colloidal suspensions is demonstrated. The films range in thickness from a monolayer to approximately fifty particle diameters depending on initial particle volume fraction. Sessile droplets of hydrogel-particle suspensions are evaporated on silicon wafers. The film is formed from particles spread densely over the air–water interface, which then cross-link and are deposited on the surface during the evaporation process. The resultant thin films exhibit a temperature-responsiveness characteristic of the individual particles permitting modulation of size, shape, porosity, and optical transmission. Recently there has been a surge in interest in using video-microscopy techniques to infer the local mechanical properties of disordered solids. One common approach is to minimize the difference between particle vibrational displacements in a local coarse-graining volume and the displacements that would result from a best-fit affine deformation. Effective moduli are then inferred under the assumption that the components of this best-fit affine deformation tensor have a Boltzmann distribution. In this paper, we combine theoretical arguments with experimental and simulation data to demonstrate that the above does not reveal information about the true elastic moduli of jammed packings and colloidal glasses. The microscopic kinetics of ubiquitous solid–solid phase transitions remain poorly understood. Here, by using single-particle-resolution video microscopy of colloidal films of diameter-tunable microspheres, we show that transitions between square and triangular lattices occur via a two-step diusive nucleation pathway involving liquid nuclei. The nucleation pathway is favoured over the direct one-step nucleation because the energy of the solid/liquid interface is lower than that between solid phases. We also observed that nucleation precursors are particle-swapping loops rather than newly generated structural defects, and that coherent and incoherent facets of the evolving nuclei exhibit dierent energies and growth rates that can markedly alter the nucleation kinetics. Our findings suggest that an intermediate liquid should exist in the nucleation processes of solid–solid transitions of most metals and alloys, and provide guidance for better control of the kinetics of the transition and for future refinements of solid–solid transition theory.
Bibliography Type:	Description: (Last Updated: 06/16/2025)

Articles in Peer-reviewed Journals	Davidson ZS, Kang L, Jeong J, Still T, Collings PJ, Lubensky TC, Yodh AG. "Chiral structures and defects of lyotropic chromonic liquid crystals induced by saddle-splay elasticity." Phys Rev E Stat Nonlin Soft Matter Phys. 2015 May;91(5):050501. Erratum in: Phys Rev E Stat Nonlin Soft Matter Phys. 2015 Jul;92(1):019905. PubMed PMID: 26066106 ; http://dx.doi.org/10.1103/PhysRevE.91.050501 , May-2015
Articles in Peer-reviewed Journals	Jeong J, Gross A, Wei WS, Tu F, Lee D, Collings PJ, Yodh AG. "Liquid crystal Janus emulsion droplets: preparation, tumbling, and swimming." Soft Matter. 2015 Sep 14;11(34):6747-54. Epub 2015 Jul 14. http://dx.doi.org/10.1039/c5sm01053e ; PubMed PMID: 26171829 , Sep-2015
Articles in Peer-reviewed Journals	Still T, Yunker PJ, Hanson K, Davidson ZS, Lohr MA, Aptowicz KB, Yodh AG. "Hydrogels: Temperature-sensitive hydrogel-particle films from evaporating drops." Advanced Materials Interfaces. 2015 Nov 9;2(16):1500371. http://dx.doi.org/10.1002/admi.201500371 , Nov-2015
Articles in Peer-reviewed Journals	Sussman DM, Schoenholz SS, Xu Y, Still T, Yodh AG, Liu AJ. "Strain fluctuations and elastic moduli in disordered solids." Phys Rev E Stat Nonlin Soft Matter Phys. 2015 Aug;92(2):022307. PubMed PMID: 26382406 ; http://dx.doi.org/10.1103/PhysRevE.92.022307 , Aug-2015
Articles in Peer-reviewed Journals	Peng Y, Wang F, Wang Z, Alsayed AM, Zhang Z, Yodh AG, Han Y. "Two-step nucleation mechanism in solid-solid phase transitions." Nature Materials. 2015 Jan;14(1):101-8. Epub 2014 Sep 14. http://dx.doi.org/10.1038/nmat4083 ; PubMed PMID: 25218059 , Jan-2015
Journal/Magazine covers	Still T, Yunker PJ, Hanson K, Davidson ZS, Lohr MA, Aptowicz KB, Yodh AG. "Cover in journal Advanced Materials Interfaces for article, 'Hydrogels: Temperature-sensitive hydrogel-particle films from evaporating drops.' " Advanced Materials Interfaces. 2015 Nov 9;2(16):1500371. Cover image: http://dx.doi.org/10.1002/admi.201570078 , Nov-2015